



Week 04 : CSS Basics (1)

What is CSS?

Have you noticed how boring simple HTML code looks?

CSS (Cascading Style Sheets) is used to control the style and layout of web pages. It defines how HTML elements should appear on the screen, in print, or on other media. CSS helps to create visually engaging websites with different styles for fonts, colours, spacing, and positioning.

For example, using CSS, you can change the background colour, font size, and position of elements on the page.

Example: The following CSS changes the background colour of a page to light blue and sets the text colour to dark blue.

```
<style>
  body {
    background-color: #ADD8E6;
    color: #00008B;
  }
</style>
```

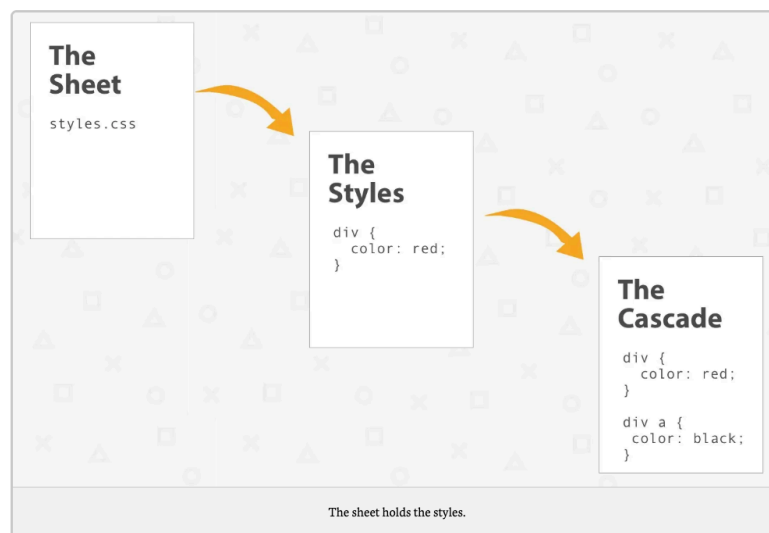


Image Source :<https://css-tricks.com/css-basics-second-s-css/>

Linking CSS to HTML

There are three ways to link CSS to HTML, each with its own benefits and use cases:

- **Inline CSS:** This method involves applying CSS directly to an HTML element using the `style` attribute. It is useful for quick, one-off styles, but it's not efficient for styling multiple elements.
- **Internal CSS:** CSS is placed within the `<style>` tag in the `<head>` section of the HTML document. It's suitable for small projects or single-page documents where external files might be unnecessary.

- **External CSS:** This method links to an external .css file using the `<link>` tag. It is the most efficient way to manage styles across multiple pages and is preferred for larger websites. Example of `<link>` tag:

```
<link rel="stylesheet" href="styles.css">
```

Priority of CSS (Cascading Order)

The priority of CSS rules is determined by the order in which they appear. The rule of "cascading" means that if multiple rules apply to an element, the last one takes precedence. The priority order from lowest to highest is as follows:

1. **External CSS (lowest priority):** External styles have the lowest priority. They can be overridden by both inline and internal CSS.
2. **Internal CSS (middle priority):** Internal styles are placed in the `<style>` tag within the `<head>` section. These styles can override external styles, but they can be overridden by inline styles.
3. **Inline CSS (highest priority):** Inline styles, defined within the HTML element using the `style` attribute, have the highest priority and override both internal and external styles.

When to Use Each Type

Inline CSS is best used for small, unique styling that applies to a single element. It's quick but not scalable for larger projects.

Internal CSS is suitable for smaller projects or single-page websites where you don't want to manage external files, and styles only apply to that specific page.

External CSS is ideal for larger websites, allowing you to centralise styles in one location. It reduces redundancy, as multiple pages can reference the same stylesheet, making it easier to maintain consistency across the site.

Example: Below is an example of internal CSS, where the styles are placed inside the `<style>` tag in the `<head>` section.

```
<html>
<head>
  <style>
    body {
      background-color: lightgray;
      font-family: Arial, sans-serif;
    }
  </style>
</head>
<body>
  <h1>Hello World</h1>
</body>
</html>
```

CSS Selectors

CSS selectors are patterns used to select and style the elements you want to target on a web page. Here are some common types of selectors:

- **Universal Selector (`*`):** Selects all elements on the page.
- **Type Selector (`element`):** Selects all elements of a specific type (e.g., `div`, `p`, `h1`).
- **Class Selector (`.classname`):** Selects elements with a specific class.
- **ID Selector (`#idname`):** Selects a specific element with a unique ID.
- **Attribute Selector (`[attribute]`):** Selects elements with a specific attribute.

```
a[href*="example.com"] {
  color: red;
```

```
font-weight: bold;
}
```

Example: The following CSS targets all `<h1>` elements, elements with the class `highlight`, and an element with the ID `main-heading`.

```
<style>
  h1 {
    color: darkblue;
  }
  .highlight {
    background-color: yellow;
  }
  #main-heading {
    font-size: 2em;
  }
</style>
```

Watch this [video](https://www.youtube.com/watch?v=cqTNHMipvhM)  [\(https://www.youtube.com/watch?v=cqTNHMipvhM\)](https://www.youtube.com/watch?v=cqTNHMipvhM) to see the CSS linking in practice

